



EDUCATION

PROGRAM	INSTITUTION	% / CGPA	COMPLETION
M.S (Thermal)	Indian Institute of Technology Madras	8.21	Jul-2022 - Ongoing
B. Tech	Dream Institute of Technology, Kolkata	8.57	Aug-2016
Class XII	Nala Inter College Nala, Jamtara	78.4	Aug-2012
Class X	R.K High School, Jamtara	79.8	Jul-2010

EDUCATIONAL ACHIEVEMENTS

- Achieved **98.51** percentile with a **GATE Score 712** (Top **1.5%** of 120,594 candidates) in Mechanical Engineering, 2021.
- Qualified for the **National Means-cum-Merit Scholarship** in 9th grade.
- Qualified for the **Rajya Medha Chatrabriti Scholarship** in 8th grade.

RELEVANT COURSEWORK

- Statistics for Machine Learning
- Numerical Methods in Thermal Engineering
- Machine Learning for Engineering and Science Application
- Inverse Methods in Heat Transfer
- Incompressible fluid flow

PROFESSIONAL EXPERIENCE

Master's Research (Jul-2022 - Ongoing)

- Understanding **vortex ring** evolution dynamics using **high-speed flow visualization** in a transparent nozzle.
- Conceptualised **end-to-end data-driven** framework to determine circulations from recorded video of vortex ring evolution.
- Investigating the effect of variations in **nozzle geometry** and internal **surface roughness** on the dynamic properties of vortex rings.

Internship, Data Science Intern (Remote) — Anti-Cheating Mechanism [My Analytics School] (Jun-2024 - Jul-2024)

- Trained a **YOLOv4** model with **93%** accuracy in detecting **unauthorized objects**, showcasing advanced **computer vision** capabilities.
- Performed detailed **data annotations**, leading to significant improvements in model accuracy, reliability, and overall prediction quality.
- Enhanced webcam images using **MIRNet** to remove noise and increase brightness, boosting model performance and image quality.

Graduate Engineer Trainee, KIC Metalicks Ltd (Jul-2019 - Jul-2020)

- Developed and **enhanced shift strategies**, improving plant operations through efficient maintenance and cross-team collaboration.
- Directed **cross-functional teamwork** for smooth plant shutdowns, minimizing downtime and ensuring timely task completion.

SKILLS

Programming Languages: Python, SQL Libraries: NumPy, Pandas Data structure & Algorithm, AI Prompting: Strategic Query

COURSE PROJECTS

Capstone Project: Data-Driven Real Estate Price Prediction, [Machine Learning (Self Project)] (Jul-2024 - Aug-2024)

- Conducted detailed **EDA** focusing on **outlier detection** and **feature selection**, improving data quality and improving model performance.
- Evaluated models, including **Linear Regression, SVR, Decision Trees, Random Forest, and XGBoost**, to determine the optimal model.
- Developed a **recommendation system** for refining real estate pricing strategies, achieving an **R² score of 0.86 with XGBoost**.

Duplicate Question Pair Identification and Email Spam Classifier, [NLP] (May-2024 - Jun2024)

- Applied NLP techniques like **vectorization, stop word removal, stemming, and lemmatization** for effective text preprocessing.
- Integrated **token, length, and fuzzy features** for duplicate question pair detection, significantly improving overall model performance.
- Built **Random Forest** and **Naive Bayes** models for question pair and email classification, scoring **0.78** and **0.84** on the **accuracy score**.

Gradient Descent and Classification Project, [Machine Learning with custom class] (Jan-2023 - Feb-2023)

- Developed custom **Python-based gradient descent** algorithms from scratch, including **batch, stochastic, and mini-batch** approaches.
- Achieved an **R² score of 0.88** on the Concrete Compressive Strength dataset through effective model hyperparameter tuning.
- Constructed a **multiclass classification** on the **Iris dataset** using **custom gradient descent**, with an **accuracy score of 0.89**.
- Cross-validated model performance by comparing results with **scikit-learn** implementations to ensure both accuracy and consistency.

Concrete Compressive Strength Prediction, [Deep Learning Project, ANN] (Mar-2023 - Apr-2023)

- Coded a neural network** from scratch, implementing forward pass, gradient calculation, and weight updates via backpropagation.
- Improved **R² by 7%** from the baseline model through hyperparameter tuning with **Grid Search CV**, enhancing predictive performance.

POSITIONS OF RESPONSIBILITY

Deputy Placement Coordinator, Placement & Internship Cell – IIT Madras (May-2023 - May-2024)

Directed placement by **pitching emails**, managing recruitment events, and **collaborating with HR**. Developed communication, provided student support, and built lasting professional relationships.

Teaching Assistant (Thermodynamics (ME1100)) – IIT Madras (Jul-2023 - Nov-2023)

Supported **problem-solving** and **concept explanations** to students while supervising the classroom environment.

HOBBIES AND EXTRA-CURRICULAR ACTIVITES

Music Enthusiast: Deeply passionate about singing & composing songs. **Storytelling:** Enjoys crafting and sharing engaging narratives. **Yoga and Meditation:** Committed to regular practice for physical and mental well-being.